

Contact

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Top Skills

Predictive Modeling

Logistics Analysis

Data Mining

Certifications

Fundamentals of MCP

Improving Deep Neural Networks:
Hyperparameter Tuning,
Regularization and Optimization

Convolutional Neural Network

Neural Networks and deep learning

AI Agents Fundamentals

Masoud Ahangary

Machine Learning Engineer | Decision Intelligence, Optimization & Scientific Computing

Rasht, Gilan Province, Iran

Summary

I work at the intersection of machine learning, applied mathematics, optimization, and scientific computing. My academic background is in Structural Engineering, with graduate research focused on the reliability analysis of gravity dams using probabilistic modeling and nonlinear finite element methods. That training shaped the way I approach problems: structurally, quantitatively, and with a strong focus on uncertainty. Over time, I transitioned into machine learning and data-driven problem solving through a range of technically demanding projects. My experience includes porting a nonlinear dimensionality reduction method (Elastic Map) from MATLAB to R, large-scale document clustering on more than 800,000 PubMed abstracts using weighted word co-occurrence networks and community detection, survival analysis on medical data, and deep learning for breast cancer classification using a hybrid LSTM-CNN architecture. For the past two years, I have been working on logistics decision systems for a used-oil collection business with multiple drivers, depots, and more than 500 customers. My work focuses on scheduling, routing, prioritization, and building analytical systems that support better operational decisions under real-world constraints. Alongside my technical work, I teach university-level mathematics privately, including algebra, calculus, probability and statistics, supervised and unsupervised learning, and signal processing. Teaching has strengthened my ability not only to solve complex problems, but also to explain them clearly and build structured understanding. I am especially interested in machine learning systems, decision intelligence, optimization, network analysis, and applied analytics for complex real-world problems.

Experience

PETRO IMEN SHARIF

Machine Learning Engineer – Logistics Decision Systems

July 2023 - Present (3 years)

Tehran

Developing data-driven decision support solutions for a used-oil collection business with multiple drivers, depots, and 500+ customers.

Designing and improving scheduling and routing logic under real-world constraints such as time windows, vehicle capacity, multi-depot operations, and driver availability.

Working on customer prioritization and planning approaches based on operational urgency, demand patterns, and service feasibility.

Translating operational challenges into structured machine learning, optimization, and analytics problems.

Supporting the development of scalable planning workflows that improve efficiency, consistency, and decision quality in daily operations.

Focusing on practical, explainable, and business-oriented analytical systems for complex logistics environments.

کارلنسر (Karlancer)

7 years 4 months

Deep Learning Researcher

February 2023 - April 2025 (2 years 3 months)

Writing a new deep algorithm class based on the random forest to predict time-history data

Machine Learning Researcher

January 2018 - April 2025 (7 years 4 months)

Freelance

Network Analyst

January 2019 - September 2022 (3 years 9 months)

Leicester City, England, United Kingdom

Freelance

Data Scientist

July 2019 - February 2020 (8 months)

Leicester City, England, United Kingdom

Writing Search-Engine Web Application For PubMed Articles to find similar articles based on documents that clustered with graph-based approach and community detection

Education

Sharif University of Technology

Master's degree, Structural Engineering · (2013 - 2016)